Roll No. :

END SEMESTER EXAMINATION DEC 2024

Name of the Course: M. Tech. (CSE)
Name of the Paper: Networking Protocols
Time: 3 Hours

Semester: I
Paper Code: MCS-104
Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer ***any two sub questions*** among a, b & c in each main question
(iii) Total marks for each main question are 20.
(iv) Each sub-question carries 10 marks

Q1	(20 marks)	
(a)	What do you understand by networking topology? Provide the figures and explain the advantages and limitations of all possible network topologies.	CO1
(b)	What is the difference between ISO-OSI Protocol stack and TCP/IP Protocol suit? Provide layered diagram for each of these. Further, show the protocols that are available at each layer of the TCP/IP protocol layer	
(c)	Consider a file of size 100 MBytes is being transferred from Computer A to Computer B over the networks that operates at (a) 56 Kbps, (b) 1 Mbps, (c) 10 Mbps and (d) 50 Mbps. Calculate the time required to transfer the file for each of the network connections	
Q2	(20 marks)	
(a)	(i) With the help of neat figures explain the idea of sharing the network address using Transparent routers and Proxy ARP. (ii) In a classful subnet addressing scheme consider an IP address 191.18.20.1 with a mask 255.255.248.0 and answer the following questions. • What is the class of IP address? • Demonstrate the usage of Mask to arrive at the logical address. Show the working in binary and decimal number systems. • What is subnet base address and host address? • What is broadcast address? • How many subnets are possible?	CO2
(b)	What are the three services provided by BOOTP? Explain. Further, with the help of a neat figure explain the role of various fields of BOOTP.	
(c)	Consider a machine that has MAC address and would like to know the IP address that it should use. Which **third generation protocol** should the machine use for this purpose? Discuss the dialogues exchanged by user and server depicting steps involved and typical **contents** of each steps/dialogue involved in address acquisition.	
Q3	(20 marks)	
(a)	(i) How do you think IPv6 improves the limitations of IPv4? Explain with the list of features (ii) What benefit does an extension header provide and how does Fragmentation and Reassembly work in IPv6? Explain each of these with the help of neat figures/headers.	CO3
(b)	Consider a classful IP address 129.2.0.0 allocated to an organization Servocon. • What is the class of IP address? Justify • This organization has 20 departments. Each of these departments require to connect at least 800 computers on its subnet. Propose IP address allocation strategy with details like, number of subnet bits, host bits, subnet mask, first and last valid host IP address for atleast five departments. Comment on the wastage/optimal utilization of IP addresses in the above case	

(c)	An ISP Telnor is assigned with the block of IP addresses 129.14.192.0/18. Telnor is approached by three organizations ORG1 to ORG3 for providing IP addresses. ORG1 is given 50%, ORG2 and ORG3 are each given 25% of IPs. (Note that Telnor also has to keep a block of IP addressed with itself for further assignments.) Now answer the following • Propose the design mentioning the IP addresses allocated to each of the three organization in Decimal and Binary notations. • For each of the Organizations provide the first and the last valid host IP address	
Q4	(20 marks)	
(a)	Consider an organization named PeopleSoft located at various geographical locations. Provide an analysis about your view on • Private Networking connecting the branch offices its advantages and limitations • Hybrid Networking, its advantages and limitations • What is the solution to above challenges? For each of the above analysis should be supported with typical networking diagram having IP addresses at each of the active devices	CO4
(b)	(i) Propose a schematic diagram for VPN connecting remote access to client by initiating a dial-up connection to a local ISP, showing all the components with proper explanation. (ii) Further, elaborate on techniques adopted by VPN for tunneling and details on encryption.	
(c)	Consider a case where multiple employees of an organization are generating a request to google.co.in using their individual computers at the same time (concurrently). The organization uses NAT having only a single public IP. Provide a solution how NAT can provide seamless communication between Google server and employee machines with the help of a neat diagram showing IP addresses, request, reply and other details.	
Q5	(20 marks)	
(a)	(i) Elaborate on the concept of domain name using labels with typical examples of levels in domain name. (ii) Provide the rules/restrictions on labels proposed by various RFCs. Further, Tabulate some of the top level domains and to whom they are assigned to.	CO5
(b)	(i) In addition to RTP and RTCP what features/ aspects are needed enabling IP telephony to interface with conventional telephones? Name the two competing agencies and their proposed suit of protocols. (ii) Draw the diagram of H.323 Protocols stack showing which one use UDP and TCP and explain role/purpose of each of the protocol.	
(c)	(i) What do you think were the VoIP driving forces and challenges involved, including encoding/digitization of the analog data. (ii) What do you understand by Jitter? Explain. Propose the methodology that can be adopted for overcoming Jitter with tradeoffs.	

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